Midterm - Jie Wen

1

(1)

Using the standard bond $PV = \sum_{t=1}^{5} \frac{C}{(1+r)^t} + \frac{1000}{(1+r)^5}$ with $r = 0.04$:

- B_A (coupon \$C= 50): **\$1,044.5182**
- B_B (coupon \$C= 80): **\$1,178.0729**

(2)

Solve Price $= \sum_{t=1}^{5} \frac{C}{(1+y)^t} + \frac{1000}{(1+y)^5}$.

- Given B_A price = \$947.7016 → **YTM = 6.2500%**
- Given B_B price = \$942.4044 → **YTM = 9.5000%**

(3)

The equity investor should prefer **company A**.
The two companies are in the same industry sector with similar current and forecasted revenues, but company B's bond has a higher YTM (9.5000% vs. 6.2500% for company A), indicating higher perceived credit risk for company B.
Given the flat risk-free rate of 4.0000%, company B has a larger credit spread (5.5000% vs. 2.2500% for company A), suggesting company A is the less risky investment.

2

For a payer-fixed / receiver-float swap:

$$\text{PV} = N \sum_{i=1}^{4} (R_{F,i} - R_S) \Delta DF_i$$

Date (mo)	t_i (yr)	OIS (r_i)	$DF_i = e^{-r_i t_i}$
2	2/12	3.75%	0.9937695
8	8/12	4.00%	0.9736857
14	14/12	4.25%	0.9516259
20	20/12	4.25%	0.9316171

Term-by-term (in $M, N = 100, \$\Delta = 0.5$):
$\text{PV}_1 = 100(0.0425 - 0.045)0.5(0.9937695) = -0.1242$
$\text{PV}_2 = 100(0.0475 - 0.045)0.5(0.9736857) = +0.1217$
$\text{PV}_3 = 100(0.0500 - 0.045)0.5(0.9516259) = +0.2379$
$\text{PV}_4 = 100(0.0550 - 0.045)0.5(0.9316171) = +0.4658$

Sum PV (payer-fixed): $+0.7012$ M

The value to the swap holder who pays fixed is $\mathbf{0.7012}$ M.

3

(1)

$\mathrm{PV}(\text{divs}) = \sum_{k=0}^{9} 8e^{-0.045(0.5+k)} = 64.4162,$
$F_A = (100 - 64.4162)e^{0.045\cdot 10} = \mathbf{55.8065}.$

(2)

$F_B = 100e^{(0.045-0.08)\cdot 10} = \mathbf{70.4688}.$

(3)

- Cash leg PV:
  $\mathrm{PV} = 8e^{-0.045\cdot 0.5} + 8e^{-0.045\cdot 1.5} = 15.2998.$
- Proportional leg: eight payments of $\alpha = 8\% \Rightarrow \prod(1-\alpha) = (0.92)^8 = 0.5132188731.$

$F_C = \Big(100 - 15.2998\Big)(0.92)^8 e^{0.045\cdot 10} = \mathbf{68.1741}.$

4

(1)

$Ke^{-rT} = 101e^{-0.04\times 2} = 101e^{-0.08} = 93.2348$
$p = 18.5 - 100 + 93.2348 = \mathbf{11.7348}$

(2)

From parity:

$$-p = -c + S_0 - Ke^{-rT}$$

So a **short put** is replicated by the static portfolio:

- **+1 share** of the stock (long S)
- **–1 European call** with the same (K, T) (short c)
- **Borrow** the present value of the strike (**short a zero-coupon bond** that pays K at T): amount $Ke^{-rT} = 93.2348$

5

(1)

- Given:
  - Initial stock price $S_0 = 90$
  - Strike price $K = 92$
  - Time to maturity $T = 1$ year
  - Number of steps $N = 2$
  - Risk-free rate $r = 5\%$ (continuous)
  - Volatility $\sigma = 35\%$
- The duration of each step is $\Delta t = T/N = 1/2 = 0.5$.
- The factors for the stock price movement are:
  - Up-factor: $u = e^{\sigma\sqrt{\Delta t}} = e^{0.35\sqrt{0.5}} \approx 1.2808$
  - Down-factor: $d = 1/u \approx 0.7808$

- The risk-neutral probability of an up-move is:
  - $p = \dfrac{e^{r\Delta t} - d}{u - d} = \dfrac{e^{0.05 \cdot 0.5} - 0.7808}{1.2808 - 0.7808} \approx 0.4891$
- The discount factor per step is:
  - $DF = e^{-r\Delta t} = e^{-0.05 \cdot 0.5} \approx 0.9753$
- $S_{uu} = S_0 u^2 = 90 \cdot (1.2808)^2 \approx 147.6411$
- $S_{ud} = S_0 ud = 90 \cdot 1.2808 \cdot 0.7808 \approx 90.0000$
- $S_{dd} = S_0 d^2 = 90 \cdot (0.7808)^2 \approx 54.8628$
- The corresponding terminal put option payoffs are calculated as $\max(K - S_T, 0)$:
  - $P_{uu} = \max(92 - 147.6411, 0) = 0.0000$
  - $P_{ud} = \max(92 - 90.0000, 0) = 2.0000$
  - $P_{dd} = \max(92 - 54.8628, 0) = 37.1372$
- Discount the expected future payoffs back to $t = 0$:
  - Value at the up-node at $t = 0.5$:
    $P_u = DF[p \cdot P_{uu} + (1 - p) \cdot P_{ud}] = 0.9753[0.4891 \cdot 0 + (1 - 0.4891) \cdot 2.0000] \approx 0.9966$
  - Value at the down-node at $t = 0.5$:
    $P_d = DF[p \cdot P_{ud} + (1 - p) \cdot P_{dd}] = 0.9753[0.4891 \cdot 2.0000 + (1 - 0.4891) \cdot 37.1372] \approx 19.4601$
  - Value at $t = 0$:
    $P_0^{EU} = DF[p \cdot P_u + (1 - p) \cdot P_d] = 0.9753[0.4891 \cdot 0.9966 + (1 - 0.4891) \cdot 19.4601] \approx \mathbf{10.1727}$

The value of the European put option is **10.1727**.

(2)

- **At the up-node at** $t = 0.5$:
  - Stock price: $S_u = S_0 u = 90 \cdot 1.2808 = 115.2723$
  - Intrinsic value: $\max(92 - 115.2723, 0) = 0.0000$
  - Continuation value: $P_u = 0.9966$
  - Node value = $\max(0.0000, 0.9966) = 0.9966$. It is not optimal to exercise early.
- **At the down-node at** $t = 0.5$:
  - Stock price: $S_d = S_0 d = 90 \cdot 0.7808 = 70.2684$
  - Intrinsic value: $\max(92 - 70.2684, 0) = 21.7316$
  - Continuation value: $P_d = 19.4601$
  - Node value = $\max(21.7316, 19.4601) = 21.7316$. It is optimal to exercise early.
- Discount the optimal values from the $t = 0.5$ nodes back to $t = 0$.
  - Intrinsic value at $t = 0$: $\max(92 - 90, 0) = 2.0000$.
  - Continuation value:
    $P_0^{AM} = DF[p \cdot P_u + (1 - p) \cdot P_d] = 0.9753[0.4891 \cdot 0.9966 + (1 - 0.4891) \cdot 21.7316] \approx 11.3046$

The final price is the maximum of the intrinsic and continuation values at $t = 0$:

$$\max(2.0000, 11.3046) = \mathbf{11.3046}$$